



Rocket Exhaust CFD Modelling in Fluent

MSc Computational Software and Techniques of Engineering
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INTRODUCTION

A few years ago, the space industry underwent exponential growth due to the emergence of private companies. This resulted in reduced costs and increased launch frequency. However, concerns about the environmental impact of rockets have been overlooked for many years due to a lack of interest and/or concern.

Aim: Model a Liquid Hydrogen-Liquid Oxygen Rocket Exhaust using Ansys Fluent through the Troposphere

- ★ Objective 1: Model exhaust plume behaviour at different altitudes.
- ★ Objective 2: Model rocket ascent through the troposphere using parameter set

LITERATURE REVIEW

Type of simulation about rocket emission:

- Atmospheric dispersion using a meteorological model with different future scenarios.
- Exhaust plume, CFD.

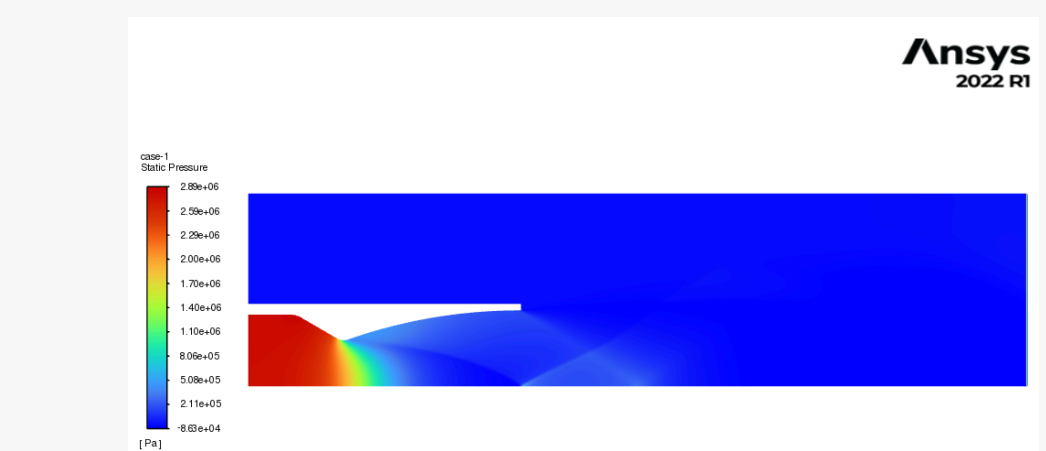
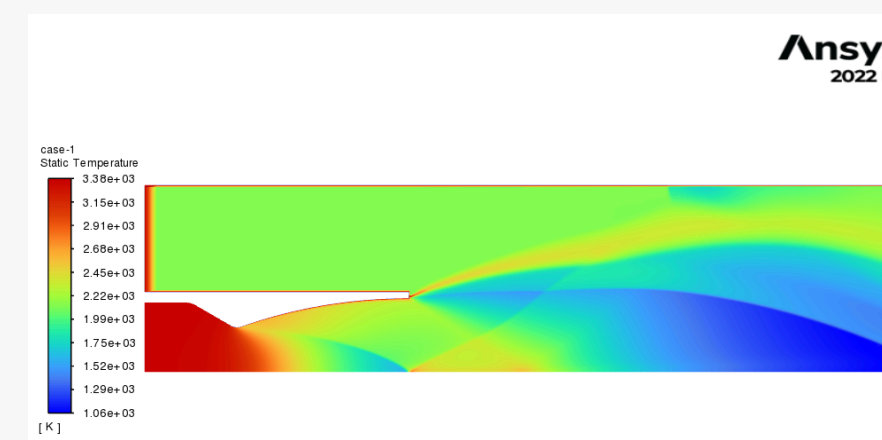
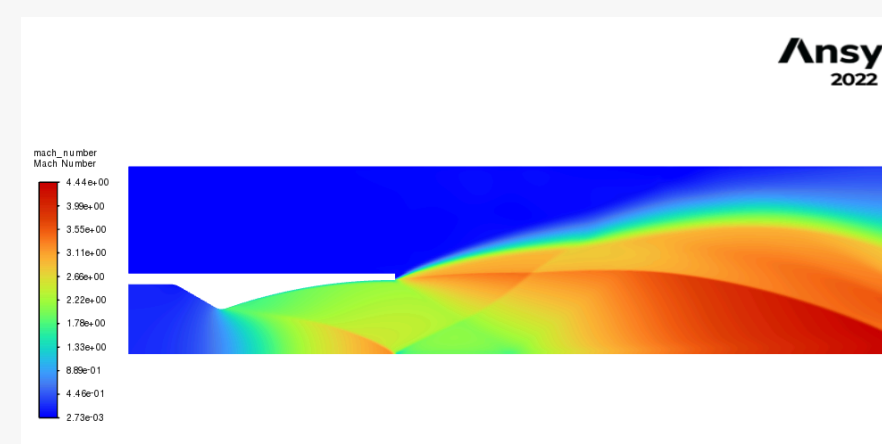
Other studies:

- Life cycle assessment of a launch vehicle
- Metal re entries of part of the rocket

Knowledge gaps: lack of studies due to the long-term undermining of its impact, and lack of regulation due to its recent exposure.

RESULTS AND DISCUSSION

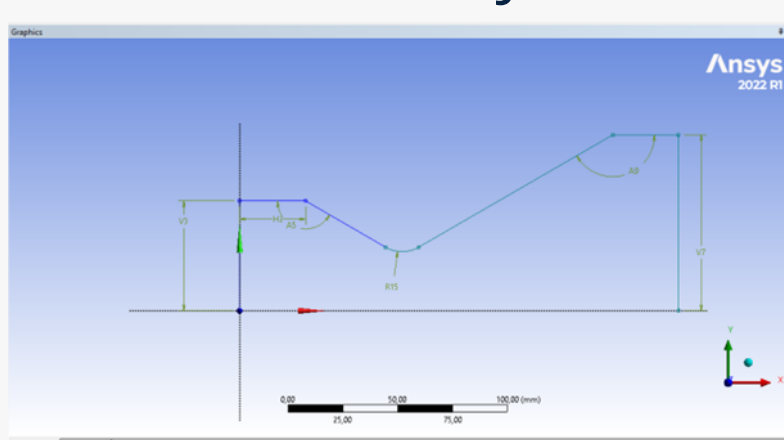
- Pressure distributions matched expected nozzle physics to generate enough thrust for the lift-off.



- Mach number matched the nozzle physics of a subsonic flux with shock and expansion waves.
- The temperature distribution showed realistic patterns everywhere except near fairfield, where applying the correct boundary conditions would cause divergence.

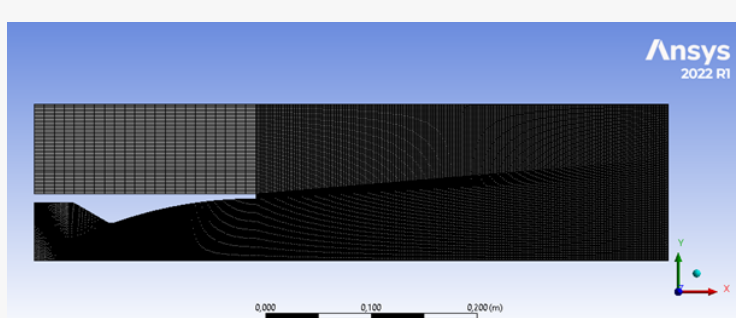
METHOD

Geometry



Engine, Throat, Nozzle and Computational domain

Mesh



+620 00 Elements

Model

- Density based
- Axisymmetric
- Viscous: k-omega SST
- Species Transport

Visualisation

- Pressure, temperature, mach number
- Species dispersion
- Mass flow rate

Boundary Conditions

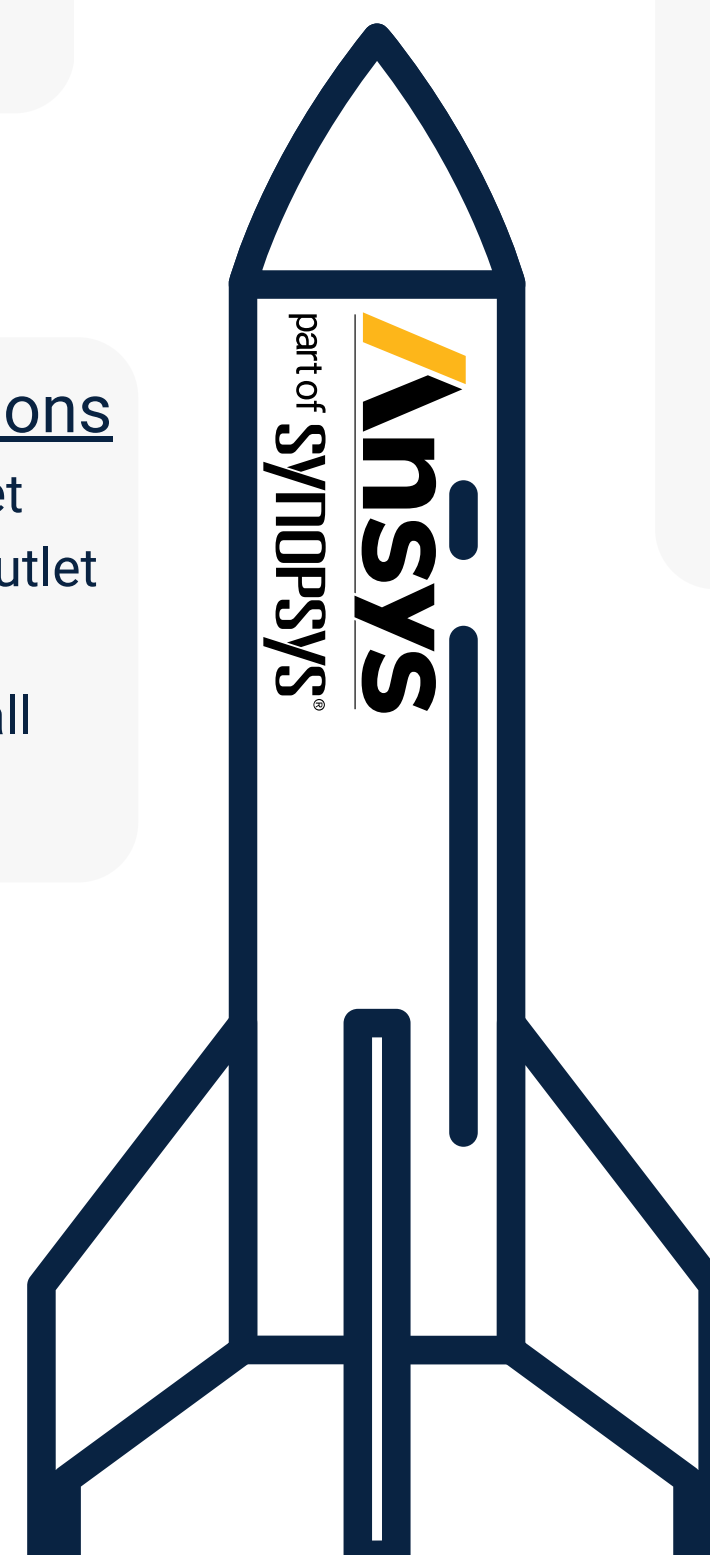
- Inlet: pressure inlet
- Outlet: pressure outlet
- Axis_sym: axis
- Boader nozzle: wall
- Fairfield: wall

Parameter sets

- Ascension troposphere

Species

$\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$
+ trace of O_2 , H_2 , O and OH



Atmospheric species VS Exhaust plume species



- Simulation through the troposphere is complex and time-consuming, both in terms of computation and cost, and achieving convergence is difficult.

CONCLUSION

- CFD modeling can reproduce key physical features but requires careful treatment of boundary conditions and species definitions.
- Results highlight the challenge of computational cost.
- Findings provide a first step toward understanding the environmental impacts of rocket launches.